

# Jonah Saitz

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## EDUCATION

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**Brown University** – Sc.B. Design Engineering, 3.9/4.0 GPA | Providence, RI | **Expected Graduation May 2026**

Relevant Courses: Control Systems Engineering, Digital Computing Systems, Linear System Analysis, Electrical Circuits and Signals

## PROFESSIONAL EXPERIENCE

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**Nextera Robotics** – Deployment Specialist | Pittsburgh, PA | November 2024 – Present

Automating construction site monitoring with AI-powered mobile industrial robots and visualization software.

- Calibrated and maintained autonomous mobile robots, capturing images of construction site progress and hazards at Carnegie Mellon University's Robotics Innovation Center.
- Collaborated with a team of engineers at a Y-Combinator-backed AI startup to troubleshoot robotic subsystems, enhance automation, LIDAR-based navigation, and environmental mapping.
- Raised sitewide image capture rate from 71% to 94%, compared to a 74% average across Nextera's other projects.

**ReMix Lab, Environmental Charter School** – Makerspace Operations Intern | Pittsburgh, PA | June 2025 – August 2025

Systematizing an academic makerspace to modernize equipment, improve safety, and reinvigorate hands-on engineering curriculum.

- Researched and authored Standard Operating Procedures (SOPs) for large inventory, including FDM 3D printers, CO2 laser cutters, and associated filtration equipment.
- Prototyped functional components such as pegboard adapters for PPE using Fusion360, encouraging makerspace safety and practical problem-solving in the classroom.
- Configured 3DPrinterOS for makerspace printers, enabling remote print management to streamline classroom workflows.
- Migrated inventory system from analog to digital, using AssetTiger and Microsoft Excel to track and visualize expenditures and resource consumption.

**Venture Outdoors** – Outdoor Educator | Pittsburgh, PA | June 2023 – August 2024

Providing experiential science education, outdoor excursions, and summer enrichment to Pittsburgh-area K-12 students.

- Taught hands-on science lessons such as electrical engineering with mini-windmills and multimeters, physics with bottle rockets, and ecology with catch-and-release fishing.
- Led kayaking, biking, and fishing excursions, helping students build teamwork, leadership, and risk management skills.

## ENGINEERING PROJECTS EXPERIENCE

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**Wilhelmus Lab, Brown University** – Undergraduate Researcher | Providence, RI | September 2025 – Present

Developing bio-inspired autonomous underwater vehicles (AUVs) to explore novel modes of propulsion and swarming capabilities.

- Design and test nitinol wire antennas to sense vortices and improve AUV navigation and tracking through feedback control.
- Build an Arduino prototype to validate vortex-sensing theory, using microcontroller, load cell amplifier, and Wheatstone bridge to track voltage fluctuations corresponding to strain gauge deflection.
- Cultivate mechatronic system design skills, prototyping antenna housing in Fusion360, fabricating with resin 3D printing, programming circuit logic, wiring and soldering connections.

**UViva, Brown University** – Electrical Lead | Providence, RI | September 2025 – Present

Protecting outdoor workers with UViva, a wearable device with accompanying software to track ultraviolet exposure and dehydration.

- Conduct surveys and interviews to gather primary data on public interest in and need for UV and dehydration tracking technology, using this market research to narrow project scope and target user group.
- Combine analog UV, capacitive touch, skin temperature, and ambient humidity sensors with an ESP32 microcontroller to calculate instantaneous UV irradiance and hydration proxy.
- Redesign basic circuit to fit inside sleek wearable housing, editing schematics and PCB layouts in KiCAD to ensure manufacturability and performance before ordering.
- Write software that visualizes integrated UV exposure and hydration data, providing employers with real-time and historical data to monitor and improve worker health and safety.

**Space Engineering, Brown University** – Avionics Team | Providence, RI | September 2022 – Present

Building a CubeSat to test the performance of experimental perovskite solar cells in the harsh orbital environment.

- Design satellite radio and power schematics for durability and long-term deployment, using KiCAD for layout.
- Test power, control, and network systems to remotely maneuver and transmit data between space and ground stations.
- Assemble, solder, and test electronics equipment for functionality and reliability in extreme environments.
- Engage K-12 students in space exploration through outreach to Providence-area schools.

## SKILLS & INTERESTS

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**Technical Skills:** Python, MATLAB, Simulink, PID control, microcontrollers (Arduino, ESP32, STM32), lab instrumentation (oscilloscope, multimeter, signal generator), KiCAD, Git/GitHub, Fusion360, 3D printing, Excel/Google Sheets

**Interests:** Engineering projects, robotics, space exploration, personal finance, outdoor recreation, weightlifting